

Experiment:9

Illustrate the concept of inter-process communication using shared memory with a C program

Code:-

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#include <unistd.h>

#include <sys/ipc.h>

#include <sys/shm.h>

#define SHM_SIZE 1024

int main() { key_t key = ftok("shmfile", 65);

if (key == -1)
{
    perror("ftok");
    exit(EXIT_FAILURE);
}
int shmid = shmget(key, SHM_SIZE, IPC_CREAT | 0666);

if (shmid == -1)
{
    perror("shmget");
    exit(EXIT_FAILURE);
}
char *shm_ptr = (char *)shmat(shmid, NULL, 0);

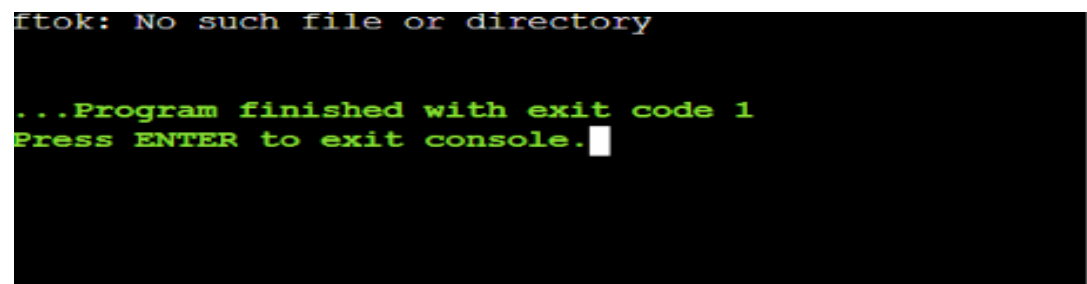
if (shm_ptr == (char *)-1)
{
    perror("shmat");
    exit(EXIT_FAILURE);
}
```

```
}
strcpy(shm_ptr, "Hello, shared memory!");

printf("Data written to shared memory: %s\n", shm_ptr);
if (shmdt(shm_ptr) == -1)
{
    perror("shmdt");
    exit(EXIT_FAILURE);
}
if (shmctl(shmid, IPC_RMID, NULL) == -1)
{
    perror("shmctl");
    exit(EXIT_FAILURE);
}

return 0;

}
```



ftok: No such file or directory

...Program finished with exit code 1

Press ENTER to exit console.

A terminal window with a black background. The first line is "ftok: No such file or directory" in red and yellow text. The second line is "...Program finished with exit code 1" in green text. The third line is "Press ENTER to exit console." in green text, followed by a white cursor box.